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The Effect of Hyper-parameters in Model-contrastive Federated Learning Algorithm

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***Abstract*—Federated learning enables all parties to perform data analysis and machine learning through the updated use of global models. The key difficulty in federation learning is data heterogeneity, and there are numerous studies that address this challenge but with poor results. The model-contrastive federated learning (MOON) algorithm described in this paper, an improvement on FedAvg, is a Model-level federation learning algorithm that is concise and efficient. The core idea is to train a local model by comparing between global models, which in turn updates the global model without the need for privacy data uploads while model-level training solves the non-IID problem. Widely tests are performed by the authors to confirm the MOON algorithm's efficacy. Contrasting various image databases, lr, μ and other different parameters reveals that MOON outperforms other federal learning algorithms in all aspects such as learning accuracy, convergence speed and training time. It demonstrates its advanced and robustness.**

Keywords—Federated learning, deep learning, model-contrastive federative learning

I. Introduction

Federated learning is a distributed machine learning technique [1], This may be broken down into federated learning that is horizontal, vertical, and federated migration learning. In horizontal federation learning, there is more feature overlap between different datasets and less user overlap. The data is divided horizontally, and the portions that have almost identical user characteristics on both sides are removed for training. As for vertical federation learning, there is more user overlap between different datasets and less user feature overlap. The data is divided into slices based on the feature dimension, and the data with nearly identical user features on both sides is removed for training. As for federated migration learning, there is less overlap between different datasets, the dataset is divided by feature dimension, and the data with nearly identical user attributes on both sides but different users is removed for training. Without communicating local data, it enables all parties to train the same model simultaneously under the control of a central server. Google first put up the idea of federated learning in 2016 [2] and utilized it to address the issue of local model updating for Android end users [3]. In practice, data is usually generated and dispersed among various parties, and most small and medium-sized enterprises face the problem of low data volume and poor quality, which is not enough to process and mine the data on its own to support the implementation of AI technology. This can lead to data owners not wanting to share their own private data with other individuals for processing. In addition, countries have introduced policies to protect data privacy and prohibit the centralization of private data to a central server for unified model training, resulting in data silos. To solve these problems, federation learning has become a research direction for many scholars, and has been widely used in the fields of medical impact, finance, natural language processing and so on.

FedAvg [4] is widely used as a classical federation learning framework in which local users are in charge of training local data to produce a local model, the server aggregates the local model to produce a global model by weighting it, and finally obtains a model that converges to the centralized machine learning result after multiple iterations. Federation learning effectively solves the dilemma of not being able to train models centrally, but a key challenge in its application is the heterogeneity of the multi-party data distribution (non-IID), which may degrade the performance of federation learning in a real-world environment where data is often distributed non-homogeneously among parties. When parties update their local models, they may move their local targets away from the global targets. As a result, the average global model will move away from the global optimal model. Solutions to this problem have been proposed, such as federated learning global model personalization, which performs additional training on local data for the purpose of adapting to individuals, however these methods require additional training, which means reduced federated learning efficiency and increased cost.

This paper presents the MOON algorithm [5]. A simple and effective federal learning framework, a modified FedAvg, changes the traditional contrast learning to compare different image data representations for model-level comparisons. The Non-IID question is solved by comparing between models.

The authors carried out extensive tests to validate the improvements of MOON, which on the CIFAR10 and CIFAR100 datasets consistently outperformed other federal learning algorithms. [6], in addition to finding optimal combinations by pairing different parameters with each other.

II. Method

A. FedAvg principle

The framework of FedAvg, a classical algorithm for federation learning, is shown in Figure 1. Each round of learning will have four steps, the parties first receive the model from the central server and download it globally.

Next, the user performs a model update via local stochastic gradient descent. After that, the updated model is then sent to the central server after the local model has been updated locally for a predetermined number of rounds. Finally, the central server accepts the parameter gradients of the local model for aggregation, and then adds the weighted average of the local model parameter gradients updating the global model using the model parameters from the previous round of aggregation as the global model to be sent to the parties in the next round.

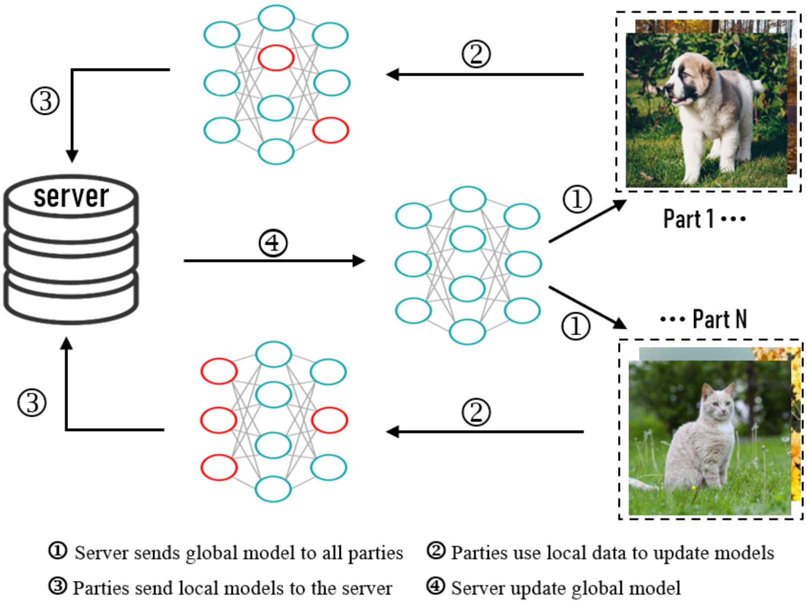

Figure 1. FedAvg framework.

B. MOON principle

MOON is an improved FedAvg. there are many studies that have improved non-IID for federation learning, which are broadly separated into two directions: the enhancement of local training (step 2 in Figure 1) and the aggregation of models by the central server (step 4 in Figure 1). the MOON algorithm improves the local training module by adding the model-contrastive loss to the local model training phase. contrastive loss. This loss consists of two components, shown in Figure 2, the typical cross-entropy loss in supervised learning and the mod-contrastive loss included in this algorithm.

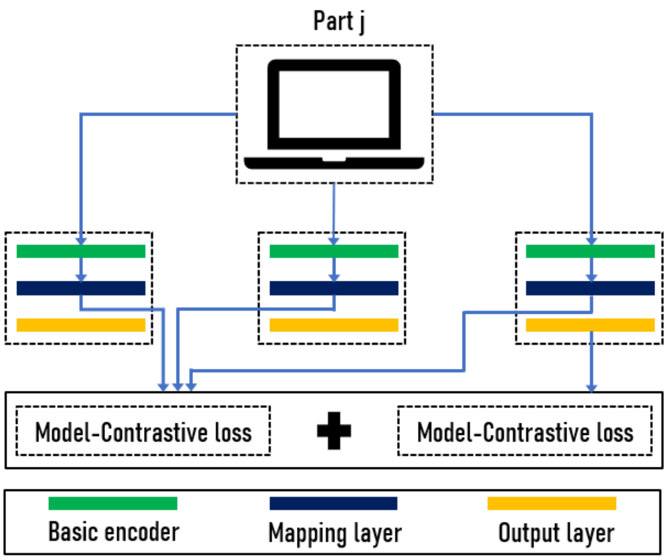

Figure 2. MOON local loss.

In addition to this, MOON takes on three representations during local training. $Z_{prev} = R_{w_i^{t-1}}(x)$, the representation obtained from the locally trained model that was sent to the central server in the previous round; $Z_{glob} = R_{w^t}(x)$, the local representation sent at the start of the round by the global model and $Z = R_{w_i^t}(x)$, the representation produced by the current round's updated local model, the goal of MOON is to keep Z close to Z_{glob}, Z far from Z_{prev}.

C. MOON method

The MOON architecture is made up of three parts: the

Basic encoder, the Mapping layer and the Output layer, which are used for extracting representation vectors from the input, mapping high-dimensional to low-dimensional, and supervised learning of the output respectively. During local training, MOON will receive the global model and update it locally. For each input, the authors will pull representations from the global model to use during the local training stage. The MOON is defined as

$$\ell_{con} = -log \frac{exp(sim(z,z_{glob})/\tau}{exp(sim(z,z_{glob})/\tau + exp(sin(z,z_{prev})/\tau} \quad (1)$$

The MOON algorithm as a whole is described as follows: the server transmits the parties' global model during every round of training and receives their locally trained model in return, after which the global model is updated using a weighted average method. Each party performs stochastic gradient descent using its local data during the local training stage in order to update the overall model.

In contrast to FedAvg, both allow each party to keep its local model and that model is constantly updated by the global model, except that MOON adds l_con to the local training and only requires each party to keep the latest model for comparison, even if it was not selected for comparison in the previous round, and when the model is decent enough, the approximation learns the same representations as the global model. Thus, while guaranteeing results, model-level comparisons solve the non-IID problem

III. Experiments

A. Experiments details

In this part, the MOON is compared with FedAvg. Two datasets are used in our experiments, including CIFAR-10 & CIFAR-100. The CNN [7] network and ResNet50 [8] network are used and the communication rounds is set up 50 for all federated learning approaches. The local epoch count is set to 5, and the number of parties is 10. To divide non-IID data among parties, pk~DirN (β) is sampled and assign a pk, j fraction of the instances of class k to party j. Dir(β) is the Dirichlet distribution with a density parameter β which is set 0.5 by default.

B. Accuracy of Different datasets

Experiments are conducted on different datasets, including cifar10 and cifar100. For cifar10, the CNN network is leveraged. For cifar100, the ResNet50 network is used because the accuracy increases too slowly on the CNN network. Figure 3 shows the accuracy between MOON and FedAvg in 50 communication rounds. As it can be observed, both on cifar10 and cifar100, the accuracy of MOON is lower than FedAvg at first. Then, as communication increases, MOON gets a better accuracy. It shows that MOON outperforms better than FedAvg on different datasets and MOON needs fewer communication rounds when it achieves the same accuracy as FedAvg.

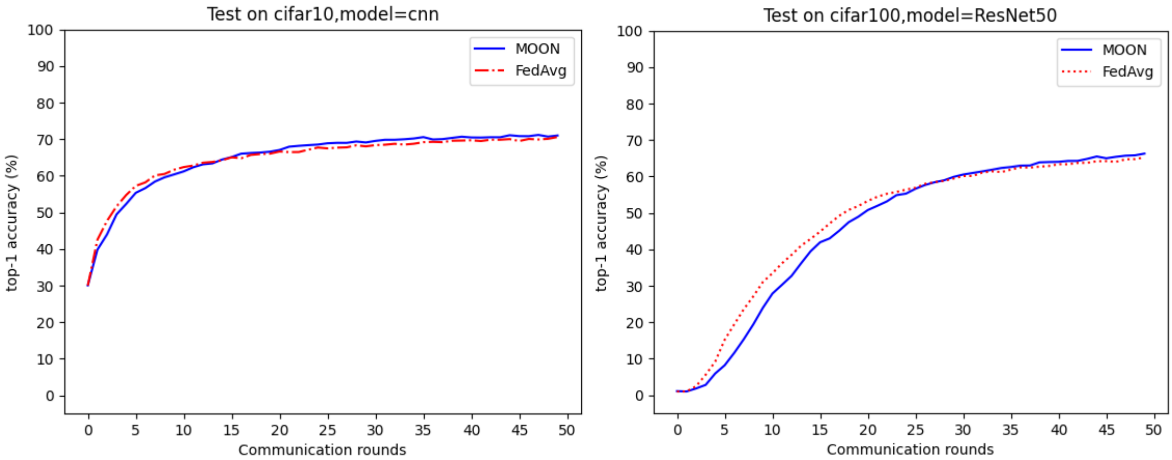

Figure 3. The global test accuracy of cifar10 and cifar100 in 50 rounds.

C. Accuracy of Different datasets

The learning rate is a hyper-parameter that regulates the neural network weights' step-by-step tuning in response to the loss gradient. The lower the gradient value, the slower the adjustment along the descent slope. This ensures that no local minima will be missed, but can also lead to slower convergence. The triangle learning rate technique enables setting an LR base as the minimum and an LR as the maximum value. The early accuracy and learning rate have a pan-direct relationship. The learning rate can achieve maximal accuracy when it reaches a critical point. Later, while the learning rate increases, the accuracy decreases. The learning rate is at its peak at this stage [9].

In this part, different learning rates is used on MOON and FedAvg. Table I shows that both algorithms are similar in that the accuracy is better at Lr = 0.01 and when the Lr is set to 0.05, the accuracy is the worst. The convergence accuracy of MOON is slightly higher than that of FedAvg when Lr is 0.01.

TABLE I The accuracy of the test with learning rate from {0.01, 0.03, 0.05}

Algorithm	Lr=0.01	Lr=0.03	Lr=0.05
MOON	71.16%	69.24%	63.82%
FedAvg	70.58%	69.97%	65.05%

D. The influence of different μ on MOON

μ is a hyper-parameter used to control the weights of the contrast loss of the model. To find out the influence of μ on the global and local test accuracy, the μ is set from {0.01,0.1,1,5}.

Figure 4 shows the global test accuracy on different μ. Under different μ, the global accuracies do not see much difference. The best accuracy is 71.17% when the μ is set to 5.

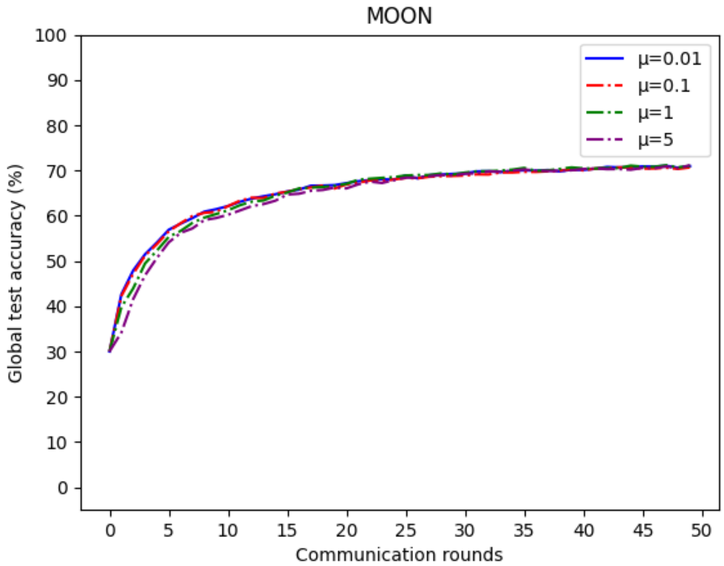

Figure 4. The Global test accuracy on μ from {0.01,0.1,1,5}.

Table II shows the local accuracy and global test accuracy on MOON and FedAvg when μ is set from {0.01,0.1,1,5}. From the results, it could be found that MOON has a higher accuracy both on global and local tests no matter what μ is. It proves that MOON performs better than FedAvg. When μ changes from 0.01 to 5, the local accuracy is getting higher and closer to the global accuracy.

TABLE II THE LOCAL AND GLOBAL TEST ACCURACY WITH μ FROM {0.01, 0.1, 1, 5}. THE MEAN ACCURACY OF 10 PARTIES IS TAKEN AS THE LOCAL ACCURACY.

Algorithm	μ=0.01	μ=0.1	μ=1	μ=5
MOON (global accuracy)	71.03%	70.67%	71.16%	71.18%
FedAvg (global accuracy)	70.58%			
MOON (local accuracy)	66.22%	66.43%	67.25%	67.43%
FedAvg (local accuracy)	66.11%			

E. The influence of different μ on MOON

Different β from {0.1, 0.5, 5} are used to generate the different non-IID data partition. It was found that the partitions were similar when the beta value was moderate and that the training accuracy of the two algorithms differed significantly when the β value was too small. As shown in Table III, MOON outperforms FedAvg for all three different β, proving that MOON is more effective and robust

TABLE III THE TEST ACCURACY WITH Β FROM {0.1, 0.5, 5}.

Algorithm	β=0.1	β=0.5	β=5
MOON	69.23%	71.16%	71.62%
FedAvg	67.34%	70.58%	70.87%

IV. DISCUSSION

In this paper, the authors modified the parameters of MOON and Fedavg to test that MOON can achieve higher learning accuracy under specific parameters, and the comparison proves that MOON has higher efficiency and better robustness in training. Firstly, the authors conducted 50 comparative experiments on Moon and Fedavg for the cifar10 dataset and CNN model, cifar100 dataset, and ResNet50 model in the experimental environment. The similarity between the two sets of experiments is that in the early stage of training, the Fedavg test accuracy is higher than that of MOON when the number of rounds is the same, but the test accuracy of MOON exceeds Fedavg in about 15 rounds and 25 rounds, respectively, which indicates that MOON has a certain potential for image processing. The difference between the two sets of experiments is that the former has a higher starting accuracy and fast convergence speed because the model is simple and requires less time per communication round. In contrast, the starting accuracy of the latter is less than 0.01, the convergence is late after a long round, and the training model is complex, resulting in a long communication round time, which affects the experimental time.

After comparing different datasets, the accuracy of the two algorithms under different learning rates LR was compared (the experimental dataset uses cifar10 The training model uses ResNet50), usually the training dataset mostly uses 0.1 or 0.01 LR. So, in the experiment, three different learning rates are between 0.1-0.01, in the process of trying LR=0.1 when the accuracy cannot converge and is not recorded in this article. It can be seen from the chart that the accuracy of Fedavg is higher than that of MOON at LR = 0.03 and LR = 0.05, but MOON is slightly higher than Fedavg at LR = 0.01, and both algorithms have higher training and testing accuracy.

Next, a comparison experiment was carried out with four different sets of μ and three different sets of β. It was found that the results of MOON were better than those of Fedavg for all variations of μ, with the highest accuracy at μ=5. The change of μ has less influence on global accuracy and greater influence on local accuracy. The greater μ is, the closer the local accuracy is to global accuracy.

In the comparison of β, the results were similar to the previous experiment, with the highest accuracy at β= 5. The balance degree of data partition changes with different β and MOON all get a b accuracy. The authors found The MOON was experimentally verified to have better results as well as robustness.

Through experiments, it is proved that MOON is better than FedAvg in different environments. However, MOON also has some problems [10]. When there are fewer communication rounds, FedAvg's accuracy is greater than MOON's. In addition, because of the addition of comparative loss, MOON needs more training time in each round.

V. CONCLUSION

Federal learning, a distributed machine learning paradigm, has the potential to break down data silos by allowing participants to model together without sharing data. When parties update their local models, their local objectives may, however, diverge significantly from the global objectives due to the heterogeneity of the data. A quick and efficient method is MOON, which is based on FedAvg. MOON proposes a model-level federated learning to solve this problem. Our experiments show that MOON improves the performance of federated learning in many aspects. MOON gets a better accuracy than FedAvg on various datasets and has a better communication efficiency. In different non-IID data experiments, MOON all get a better accuracy, which proves that MOON is more robust. The authors also test different parameters on MOON. With the increase of learning rate, MOON does not get a better accuracy. The influence of μ on precision is also tested. The authors find that the change of weight of model-contrastive loss has little impact on global accuracy. However, it has a more obvious impact on local accuracy. When the weight gets bigger, the accuracy of local model's accuracy is getting closer to the global model's accuracy. It shows that local models learn better representations than the previous one with a bigger weight of model-contrastive loss.

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